

Question number	Answer	Additional guidance	Mark
6(a)(i)	B Q is the SAN and P is the AVN A is not correct because R is not the AVN C is not correct because Q is not the bundle of His D is not correct because P is not the Purkinje fibres		(1)

Question number	Answer	Additional guidance	Mark
6(a)(ii)	B myogenic A is not correct because diastole is not the term. C is not correct because polarisation is not the term. D is not correct because systole is not the term.		(1)

Question number	Answer	Additional guidance	Mark
6(b)(i)	A explanation that includes the following points: - cardiovascular control centre detects a drop in pH of blood (or increased CO₂) (1) - therefore SAN stimulated to generate more impulses per second (1) - therefore the cardiac cycle is shorter (1)		(3)

Question number	Answer	Additional guidance	Mark
6(b)(ii)	A calculation showing the following step: Correct calculation of stroke volume	(4500/51) 88.24	(1)

Question number	Answer	Additional guidance	Mark
6(b)(iii)	A calculation showing the following steps: correct calculation of heart beat in unit time.	Eg 19 beats in 7 seconds $60/7 \times 19 = 162.8 = 163$ Do not accept fraction of heart rate. (PE to give an acceptable range.)	(1)

Question number	Answer	Additional guidance	Mark
6(b)(iv)	An answer that includes three of the following points: - comment on the rhythm of the two traces (1) - comment on the frequency of the two traces (1) - comment on the size of the R wave (1) - comment on the ST segment of the wave (1)	For each mark there must be a clear comparison between the two ecgs. Eg more normal PQRST wave in second trace Eg less heart beats in second trace Eg R wave higher in first trace Eg no clear S segment in first trace	(3)

